MULTI-VENDOR CARRIER ETHERNET & OPTICAL CPE HARDWARE MANUAL (REV 4.0)

Document ID: ENG-HW-OPT-MAN-V4 | Scope: Cisco ASR-920, Juniper ACX2100, and Nokia 7210 SAS-Mxp (NOK-7210-SAS-Mxp) Platforms

Chapter 1: General Safety, Rack Mounting & Power Supply Specifications

1.1 Purpose and Scope: This engineering manual defines the physical installation, electrical power, thermal management, copper interface, and optical SFP+/XFP transceiver operating parameters for enterprise demarcation and access aggregation routers deployed across the national Carrier Ethernet footprint. All field technicians and automated NOC assurance systems must adhere strictly to the hardware tolerances specified herein.

1.2 Laser Safety Compliance (IEC 60825-1 Class 1): Optical transceivers deployed in access and metro chassis emit invisible infrared laser radiation at 850 nm, 1310 nm, or 1550 nm wavelengths. Never inspect open fiber patch cords or LC/UPC bulkhead adapters without an approved optical power meter and attenuation filter.

1.3 Chassis Rack Mounting & Grounding: All CPE and aggregation shelves must be mounted in standard EIA-310 19-inch two-post or four-post telecom racks using dual M6 lug earthing straps bonded to the central station ground bus (< 1.0 Ohm resistance).

Chassis Platform SKU	Form Factor	Redundant PSU Input	Max Power Draw	Chassis Ground Lug
CISCO-ASR-920-10G	1 RU Fixed	-48V DC Dual Feed / 110-240V AC	145 W	Dual M6 Ring Terminal
JUNIPER-ACX-10G	1 RU Hardened	-48V DC Dual Feed	160 W	Dual M6 Ring Terminal
NOK-7210-SAS-Mxp	1.5 RU Extended	-48V DC / +24V DC Modular	185 W	Dual M6 Ring Terminal

Chapter 2: Environmental, Thermal & Fan Tray Thresholds

2.1 Ambient Operating Temperature & Humidity: Enterprise demarcation units installed in unconditioned street cabinets or customer basement MDF rooms must support extended industrial temperature envelopes (-40C to +65C for passive cooled variants; -5C to +55C for fan-cooled optical shelves) at 5% to 95% non-condensing relative humidity.

2.2 Fan Tray RPM Telemetry & Thermal Shutdown: Each chassis houses hot-swappable PWM fan trays monitored via SNMP ENTITY-SENSOR-MIB and gNMI /components/component/state/temperature. If exhaust ASIC temperature exceeds +78C, the shelf raises a MINOR thermal alarm and forces fan speed to 100% duty cycle. Thermal shutdown engages automatically at +92C to prevent silicon damage.

Platform SKU	Normal Intake Temp	Minor Thermal Alarm	Critical Shutdown	Fan Tray FRU SKU
CISCO-ASR-920-10G	0C to +45C	+72C	+88C	ASR-920-FAN-M
JUNIPER-ACX-10G	-20C to +55C	+78C	+92C	ACX2100-PASSIVE
NOK-7210-SAS-Mxp	-5C to +50C	+75C	+90C	3HE09152AA-FAN

Chapter 3: Copper RJ-45 & Electrical Interface Specifications (Cisco ASR-920 & Juniper ACX2100)

3.1 10/100/1000BASE-T RJ-45 Client Handoff Ports: For legacy sub-1G enterprise LAN handoffs, Cisco ASR-920 and Juniper ACX2100 chassis provide integrated IEEE 802.3ab auto-negotiating RJ-45 copper ports supporting Cat5e/Cat6 structured cabling up to 100 meters.

3.2 Electrical MDI/MDI-X & TDR Cable Diagnostics: Time Domain Reflectometry (TDR) can be invoked on copper ports ONLY to detect open/short pair faults. Note: Copper RJ-45 electrical parameters do NOT apply to 10GBASE-LR optical fiber circuits and must be ignored by optical fiber assurance workflows.

Platform Model	Port Range	Media Standard	Max Distance	PoE+ Support	Auto-MDIX
CISCO-ASR-920-10G	Gi0/0/0 - Gi0/0/11	1000BASE-T Cat6	100 m	No	Enabled
JUNIPER-ACX-10G	ge-0/0/0 - ge-0/0/15	1000BASE-T Cat6	100 m	802.3at (Ports 0-3)	Enabled

Chapter 4: Optical SFP+/XFP Transceiver Operating Parameters

4.1 Multi-Vendor 10GBASE-LR Optical Transceiver Overview: Enterprise 10 Gbps Carrier Ethernet circuits (including service spec ETH-10G-GOLD) terminate on 10GBASE-LR (1310 nm single-mode fiber, up to 10 km reach) SFP+/XFP optical transceivers with SFF-8472 Digital Diagnostics Monitoring (DDM) enabled. Each vendor platform enforces distinct photodiode receive sensitivity windows.

4.2 Nokia 7210 SAS-Mxp (SKU: NOK-7210-SAS-Mxp) Optical Operating Envelope & DDM Recalibration Rules: For physical CPE model SKU NOK-7210-SAS-Mxp equipped with 10GBASE-LR 1310nm optics, the authoritative optical power thresholds are: - Nominal Transmit (Tx) Power: -1.0 dBm - Receiver Overload Cutoff (Max Rx): -1.0 dBm - Normal Operating Rx Power Range: -14.0 dBm to -24.9 dBm (optimal FEC-free window) - Soft Degradation / DDM Recalibration Window: -25.0 dBm to -27.5 dBm (elevated BER/packet loss caused by photodiode AGC bias drift; eligible for non-intrusive remote TMF640 DDM optical transceiver recalibration) - Critical Loss-of-Signal (LOS) / Hard Attenuation Cutoff: < -27.5 dBm (physical macro-bend, dirty splice, or fiber cut; remote DDM recalibration is strictly ineffective and forbidden — mandatory TMF621 field fiber technician dispatch required).

Hardware Model SKU	Optical Standard	Nominal Tx	Overload Rx	Normal Rx Range	DDM Recalibration Window	Critical LOS Cutoff
CISCO-ASR-920-10G	10GBASE-LR 1310nm	+0.5 dBm	-0.5 dBm	-12.0 to -22.9 dBm	-23.0 to -25.5 dBm	< -25.5 dBm
JUNIPER-ACX-10G	10GBASE-LR 1310nm	+1.0 dBm	-0.5 dBm	-13.0 to -23.9 dBm	-24.0 to -26.5 dBm	< -26.5 dBm
NOK-7210-SAS-Mxp	**10GBASE-LR 1310nm**	**-1.0 dBm**	**-1.0 dBm**	**-14.0 to -24.9 dBm**	**-25.0 to -27.5 dBm**	**< -27.5 dBm**

Chapter 5: Transceiver Diagnostics, DDM Polling & Field Fiber Cleaning Procedures

5.1 Real-Time Singleton DDM Telemetry Probing: Cached NMS performance graphs poll every 15 minutes and must NEVER be used to authorize an optical laser/AGC recalibration. Automated assurance agents must invoke a synchronous, non-cached gNMI/REST singleton optical probe (bypass_cache=true) against the terminating OLT/CPE port prior to issuing any TMF640 service configuration action.

5.2 Field OTDR Testing & End-Face Inspection: When live optical Rx power falls strictly below -27.5 dBm on a NOK-7210-SAS-Mxp interface, dispatched field engineers must perform bidirectional OTDR traces at 1310 nm and 1625 nm and inspect LC/UPC ferrules per IEC 61300-3-35 before replacing the SFP+ module.